MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A chord of a circle of radius 10 cm subtends a right angle at the centre. The area of the minor segment is: (Use $\pi = 3.14$) [1]

(A) 28.5 cm^2 (B) 50 cm^2 (C) 78.5 cm^2 (D) 25 cm^2

Q2. In a circle of radius 14 cm, a chord subtends an angle of 90° at the centre. The area of the major segment is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 560 cm^2 (B) 588 cm^2 (C) 616 cm^2 (D) 532 cm^2

Q3. If the circumferences of two circles are in the ratio 4 : 9, then the ratio of their areas is: [1]

(A) 2 : 3 (B) 4 : 9 (C) 9 : 4 (D) 16 : 81

Q4. If the areas of two circles are in the ratio 25 : 36, then the ratio of their circumferences is: [1]

(A) 5 : 6 (B) 25 : 36 (C) 6 : 5 (D) 125 : 216

Q5. **Assertion (A):** The area of a minor segment of a circle is always strictly less than the area of the corresponding sector.
Reason (R): The area of a minor segment is obtained by subtracting the positive area of the corresponding triangle $\triangle OAB$ from the sector area. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the length of the arc of the corresponding minor sector. (Use $\pi = 3.14$) [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Three students are seated at equal distances on the boundary of a circular park of radius 20 m. Find the length of each string connecting them and the area of the equilateral triangle formed. (Use $\sqrt{3} = 1.732$) [3]

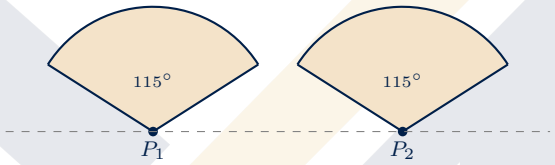
Q8. In a circle of radius 10 cm, a chord subtends a right angle at the centre. Find the area of: (i) the minor segment, and (ii) the major sector. (Use $\pi = 3.14$) [3]

Q9. In the quadrant $OPBQ$ of a circle with centre O and radius 3.5 cm, a point D lies on OB such that $OD = 2$ cm. Find the area of the shaded region. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Automobile Aerodynamic Dual Windshield Wipers

An automotive safety engineer designs a dual windshield wiper system for a passenger car. The system has two blades of length $r = 25$ cm each. Each wiper sweeps through a maximum operating angle of 115° without any overlapping region between the two blades.



(i) State the formula for the area swept by a wiper blade modeled as a sector of a circle. [1]

(ii) Calculate the area swept by a single wiper blade in one complete sweep. [1]

(iii) Calculate the total windshield surface area cleaned by both wipers simultaneously at each sweep, OR find the total arc distance traveled by the outer tip of both wiper blades. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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